EV Battery Course — Module 6: Battery Pack Design and the 60 kWh Pack Project

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A conceptual 60 kWh pack for an Indian compact SUV comes out at 120S1P LFP prismatic cells, 384 V nominal, about 61 kWh gross, about 470 kg and roughly $4,900 at the stated assumptions. This module teaches the method and then works the full project. It is a conceptual design exercise, not a validated production design.

1. Learning objectives

- Turn vehicle requirements into pack energy, voltage and power targets.
- Choose a cell and a series-parallel arrangement.
- Calculate gross and usable energy, currents, C-rates, resistive losses and voltage sag.
- Estimate pack mass, volume, cooling need and cost from stated assumptions.
- Explain the role of fuses, contactors, pre-charge, HVIL, isolation monitoring, busbars and the enclosure.

2. Fundamentals: the design chain

1. **Vehicle requirement:** range, performance, charging time, cost, package space.
2. **Energy:** usable energy = range × consumption; gross = usable ÷ usable window.
3. **Voltage architecture:** 400 V or 800 V, set by charging targets, inverter and motor, cost.
4. **Cell choice:** chemistry, format, capacity; drives N_s and N_p.
5. **Power check:** peak and continuous current within cell C-rate limits and thermal limits.
6. **Mechanical and thermal:** packing, cooling, crash, sealing, mass.
7. **Electrical protection:** fuse, contactors, pre-charge, HVIL, isolation.
8. **Cost and iteration.**

3. Detailed technical explanation

400 V vs 800 V. 400 V class is cheaper today (mature parts, silicon IGBT inverters) and fine for DC charging up to about 150 kW. 800 V supports 250–350 kW charging and thinner cables but needs silicon carbide (SiC) power electronics and higher-rated components.

Usable vs gross energy. The BMS keeps the pack away from 0% and 100% to protect life. Typical usable windows are about 90–97% of gross; the OEM decides the margin.

Continuous vs peak power. Peak (about 10 s) sets acceleration; continuous (minutes) sets hill-climbing and sustained high speed. Cells have separate peak and continuous current ratings, both temperature-dependent.

DCIR and resistive losses. Pack resistance = cell DC internal resistance × $N_s \div N_p$ + interconnects + busbars + contactors + fuse + harness. Losses scale with I^2.

Protection architecture

Component	Function
Main fuse (pyrofuse or melting fuse)	Opens on short-circuit currents the contactors cannot safely break
Main contactors (+ and −)	Connect and disconnect the pack under BMS control
Pre-charge relay + resistor	Charges the inverter's capacitor slowly before the main contactor closes, preventing inrush and contact welding
HVIL (high-voltage interlock loop)	Low-voltage loop through every HV connector; opening any connector drops the contactors
Isolation monitoring	Measures resistance between HV and chassis; flags insulation faults
Manual service disconnect	Physical break for service technicians
Current sensor (shunt or Hall)	Feeds the BMS for SOC and protection

Busbars, connectors, insulation, grounding. Copper or aluminium busbars are sized for current density and temperature rise; HV connectors are rated and keyed; creepage and clearance distances follow voltage class; the enclosure is bonded to chassis ground; HV is fully isolated from chassis (a floating system).

Enclosure. Usually aluminium extrusions or steel stampings; must survive side-pole and underbody impacts, vibration (road profiles over life), water ingress (IP67, often IP68/IP6K9K for washing and wading), and must vent cell gases safely.

4. Diagram: pack electrical architecture

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flowchart LR
  CELLS["120 cells in series"] --> FUSE["Main fuse"] --> CPLUS["Main
contactor +"] --> INV["Inverter / DC link"]
  CELLS --> SHUNT["Current sensor"] --> CMIN["Main contactor −"] --> INV
  CPLUS -. "bypassed at start-up" .- PRE["Pre-charge relay + resistor"]
  BMS["BMS: cell voltages, temps, current, isolation, HVIL"] --> CPLUS
  BMS --> CMIN
  BMS --> PRE
```

5. Step-by-step: power-up and power-down

1. Driver presses start; 12 V wakes the BMS.
2. BMS checks HVIL continuity, isolation resistance, cell voltages and temperatures.
3. Negative contactor closes; pre-charge relay closes; DC-link capacitor charges through the resistor.
4. When DC-link voltage reaches about 95% of pack voltage, the positive main contactor closes and the pre-charge relay opens.
5. Drive. BMS continuously limits charge and discharge power.
6. On shutdown or fault, the inverter reduces current first, then contactors open. In a crash, the airbag controller signals contactors (and pyrofuse) to open.

6. Engineering equations

Road-load power:

$$P_{wheel} = \left(m g C_{rr} + \tfrac{1}{2} \rho C_d A v^2\right) v$$

Energy targets:

$$E_{usable} = d \times c \qquad E_{gross} = \frac{E_{usable}}{\eta_{window}}$$

Cell arrangement:

$$N_s = \frac{V_{pack,target}}{V_{cell}} \qquad N_p = \frac{E_{gross}}{N_s \, V_{cell} \, C_{cell}}$$

Current, C-rate, losses and sag:

$$I = \frac{P}{V_{pack}} \qquad \text{C-rate} = \frac{I}{N_p C_{cell}} \qquad P_{loss} = I^2 R_{pack} \qquad \Delta V = I R_{pack}$$

Pack resistance:

$$R_{pack} = \frac{N_s R_{cell}}{N_p} + R_{interconnects} + R_{HV\,path}$$

Coolant flow:

$$\dot{m} = \frac{\dot{Q}}{c_p \, \Delta T}$$

Pre-charge:

$$\tau = R C \qquad V_C(t) = V_{pack}\left(1 - e^{-t/\tau}\right) \qquad E_R = \tfrac{1}{2} C V^2$$

7. The 60 kWh pack project

7.1 Vehicle assumptions

Parameter	Value (assumed)
Vehicle	Compact SUV, Indian market
Kerb mass with pack	1,700 kg
Rolling resistance C_{rr}	0.009
Drag C_d × frontal area A	0.32 × 2.4 m²
Air density ρ	1.2 kg/m³
Drivetrain efficiency	87%
Average auxiliary load	0.6 kW
Target real-world range	~400 km
Peak / continuous motor power	150 kW / 70 kW
DC fast charging	100 kW

7.2 Step 1 — consumption check at 90 km/h

- v = 25 m/s. Rolling force = 1,700 × 9.81 × 0.009 = 150.1 N.
- Aero force = 0.5 × 1.2 × 0.32 × 2.4 × 25² = 288.0 N.
- P_{wheel} = (150.1 + 288.0) × 25 = 10,953 W.
- $P_{battery}$ = 10,953 ÷ 0.87 + 600 = 12,590 + 600 = 13,190 W.

- Consumption = 13,190 W ÷ 90 km/h = **147 Wh/km**. Use 145 Wh/km as the mixed-use design figure.

7.3 Step 2 — energy

- E_usable = 400 × 145 = **58.0 kWh**.
- Usable window 95% → E_gross = 58.0 ÷ 0.95 = **61.1 kWh**.

7.4 Step 3 — voltage and cell selection

- Choose 400 V class (cost, 100 kW charging is enough).
- Choose LFP prismatic, 3.2 V nominal, 160 Ah (512 Wh), operating window 2.5–3.65 V, DCIR 0.25 mΩ, about 175 Wh/kg and 380 Wh/L. (Representative values, not a specific product.)
- N_s = 384 ÷ 3.2 = **120**.
- N_p = 61,100 ÷ (120 × 3.2 × 160) = 61,100 ÷ 61,440 = 0.99 → **1**.

7.5 Step 4 — pack values

Quantity	Calculation	Result
Configuration	—	**120S1P, 120 cells**
Nominal voltage	120 × 3.2	**384 V**
Voltage range	120 × 2.5 to 120 × 3.65	**300–438 V**
Capacity	1 × 160	**160 Ah**
Gross energy	384 × 160	**61.44 kWh**
Usable energy (95%)	61.44 × 0.95	**58.4 kWh**
Estimated range	58,368 ÷ 145	**~403 km**

7.6 Step 5 — currents and C-rates

Condition	Power	Current at 384 V	C-rate
Peak acceleration (10 s)	150 kW	390.6 A	2.44C
Peak at low SOC (~340 V)	150 kW	441.2 A	2.76C
Continuous	70 kW	182.3 A	1.14C

Condition	Power	Current at 384 V	C-rate
Cruise 90 km/h	~13 kW	34.3 A	0.21C
DC fast charge	100 kW	260.4 A	1.63C

Check the chosen cell's datasheet: an LFP power-capable cell rated about 3C pulse and 1–1.5C continuous fits, with temperature-dependent derating.

7.7 Step 6 — resistance and losses

Assumptions: cell DCIR 0.25 mΩ; 120 cell interconnects at 0.05 mΩ each; HV path (fuse, contactors, shunt, connectors) 4 mΩ.

- R_pack = 120 × 0.25 + 120 × 0.05 + 4 = 30 + 6 + 4 = **40 mΩ**.

Condition	Current	Loss I^2R	Sag I × R
Peak 150 kW	390.6 A	**6.10 kW** (4.1%)	15.6 V
Continuous 70 kW	182.3 A	**1.33 kW** (1.9%)	7.3 V
Cruise	34.3 A	**0.05 kW**	1.4 V
DC fast charge	260.4 A	**2.71 kW**	10.4 V

This ignores polarisation beyond DCIR and the change in resistance with temperature and SOC; cold cells may have 2–3× these values.

7.8 Step 7 — mass and volume

- Cell mass = 512 ÷ 175 = 2.93 kg → 120 cells = **351 kg**.
- Assume cell-to-pack mass ratio 75% (CTP LFP): pack mass = 351 ÷ 0.75 ≈ **468 kg** → 61,440 ÷ 468 = **131 Wh/kg at pack level**.
- Cell volume = 512 ÷ 380 = 1.35 L → 120 cells = **162 L**.
- Assume 60% volume utilisation: pack volume ≈ **270 L**, e.g. about 1,700 × 1,300 × 125 mm under the floor.

7.9 Step 8 — thermal management requirement

- Worst sustained heat: DC fast charge, about 2.7 kW I^2R plus reversible and polarisation heat; design the cooling for about **4 kW** with margin.
- Without cooling: thermal mass ≈ 351 kg × 1,000 J/(kg·K) = 351 kJ/K. A 25-minute fast charge at 2.7 kW = 4.05 MJ → ΔT ≈ **11.5 K**, too much starting at 35 °C ambient.

- Liquid cooling plate under the cells with water-glycol ($c_p \approx 3{,}500$ J/(kg·K)), allowed coolant rise 5 K: $\dot{m} = 4{,}000 \div (3{,}500 \times 5) = 0.229$ kg/s ≈ **13 L/min**.
- Plus a heater (PTC or heat-pump loop) of about 3–5 kW for cold starts and preconditioning, and a chiller linked to the AC circuit for Indian summers.

7.10 Step 9 — protection sizing (indicative)

- Contactors: rated ⩾ 450 V DC, continuous ⩾ 250 A, short peaks ⩾ 450 A.
- Pre-charge: assume DC-link C = 1,000 µF; target 95% in about 300 ms → $3\tau = 0.3$ s → $\tau = 0.1$ s → $R = 0.1 \div 0.001 =$ **100 Ω**. Inrush = 438 ÷ 100 = 4.4 A maximum. Energy in the resistor = $0.5 \times 0.001 \times 438^2 =$ **96 J** per pre-charge.
- Isolation: UNECE R100 / AIS-038 require at least 100 Ω/V for DC circuits → 438 V × 100 = **43.8 kΩ minimum**; in practice monitors warn at much higher values.

7.11 Step 10 — cost estimate

Assumptions: LFP cell price $60/kWh (consistent with 2025 LFP market levels, see Module 4); cells are 75% of pack cost; ₹88 per US$ (illustrative). Excludes duties, warranty reserve and OEM margin.

- Cells: 61.44 × 60 = **$3,686**.
- Pack: 3,686 ÷ 0.75 = **$4,915** ≈ **$80/kWh** ≈ **₹4.3 lakh**.

7.12 Limitations of this conceptual design

- Cell data are representative, not from a real datasheet; real designs use supplier data across temperature and SOC.
- No detailed thermal simulation (CFD), crash simulation or vibration analysis.
- Ageing ignored: end-of-life capacity (e.g. 80%) and resistance growth (e.g. +50%) must still meet range and power targets.
- Regulatory testing (AIS-038, AIS-156 where relevant, UNECE R100, IS 16893, the Indian adoption of IEC 62660, for cells) not included.

8. Real-world EV examples

- Tata Nexon EV and Curvv EV, MG ZS EV and Mahindra BE 6 use 400 V-class LFP packs in a similar energy range.
- Hyundai E-GMP and Porsche Taycan use 800 V NMC packs to enable very fast charging.
- BYD Blade packs illustrate high volume utilisation with long LFP cells in CTP layout.

9. Manufacturing applications

- Fewer, larger cells (1P configuration) reduce weld count and BMS channels, but every cell is a single point of failure; cell quality must be excellent.
- Torque-controlled fasteners and laser welds on busbars are critical-to-quality; record each joint's resistance for traceability.
- End-of-line tests: insulation resistance, HVIL, contactor function, pre-charge timing, BMS communication, leak test of enclosure and coolant circuit.
- Should-cost: cells about 70–80% of pack cost; then enclosure, thermal system, BMS and electronics, HV components, assembly labour and test.

10. Common mistakes

- Designing for nominal voltage only; components must handle the full voltage range (here 300–438 V).
- Forgetting end-of-life: a pack that just meets range when new will fail customers after ageing.
- Ignoring cold-temperature power limits.
- Treating pack Wh/kg as cell Wh/kg; the enclosure, cooling and electronics add 25–40% mass.
- Using peak C-rate for continuous design.

11. Practical case study: contactor welding at a fleet operator

Problem. Several taxis fail to start; diagnostics show welded main contactors.

Root cause. A software update shortened the pre-charge wait; in hot weather the pack voltage and DC-link voltage differed by more than designed at closure, causing an arc and contact welding.

Correction. Restore closure condition (DC-link ≥ 95% of pack voltage, not a fixed time), add a pre-charge timeout fault, and inspect contactors in affected vehicles.

Lesson. Protection logic is as important as hardware ratings.

12. Revision notes

- Range × consumption → usable → gross; voltage target → N_s; energy → N_p.
- Check currents at peak, continuous and charging against cell limits.
- R_pack and I^2R give losses and heat; size cooling for the worst sustained case.
- Pack mass ≈ cell mass ÷ 0.65–0.8.

- Pre-charge, HVIL, isolation monitoring and fuses are mandatory safety features.

13. Glossary

Term	Meaning
DC link	Capacitor bank at the inverter input
Pre-charge	Controlled charging of the DC link before main contactor closure
HVIL	High-voltage interlock loop
Pyrofuse	Explosively actuated disconnect
IP67	Dust-tight; immersion to 1 m for 30 min
Usable window	Fraction of gross energy the BMS allows
DCIR	DC internal resistance, from a current pulse

14. Knowledge check

Conceptual

1. Why must the BMS close contactors in a particular sequence?
2. Why is end-of-life data more important than beginning-of-life data in pack sizing?
3. Why might an OEM choose 1P large cells vs 3P smaller cells?
4. What does isolation monitoring protect against?
5. Why does pack Wh/kg fall well below cell Wh/kg?

Numerical

1. Redesign the pack as 800 V class with the same cell: N_s and N_p for about 61 kWh? What happens to cell capacity needed?
2. For the 120S1P pack, compute losses at 120 kW peak.
3. At end of life, capacity falls to 80% and resistance rises 50%. Find range and peak-power loss at 150 kW.

Application

1. Your costing team receives a quote of $95/kWh for a pack. Using this module's structure, where would you challenge it?
2. An Indian OEM wants 30-minute 10–80% charging on this pack. Is 100 kW enough?

15. Answer key

Conceptual

1. To avoid inrush current welding the contactors and damaging the DC link; pre-charge first, then main contactor, then open the pre-charge relay.
2. Customers and warranties judge range and power over the vehicle's life; the pack must still meet targets after capacity fade and resistance growth.
3. 1P: fewer parts, welds and BMS channels, lower cost. 3P: a weak cell is supported by its neighbours, and cells are standard sizes. Trade-off between simplicity and fault tolerance.
4. Insulation breakdown between HV and chassis, which could expose people to shock if a second fault occurs.
5. Enclosure, cooling plates, busbars, BMS, HV components and fixings add mass that stores no energy.

Numerical

1. 768 V → N_s = 240 at 3.2 V. 61,440 ÷ (240 × 3.2) = 80 Ah per series position, so either an 80 Ah cell in 240S1P or a 40 Ah cell in 240S2P. Current halves; cell count stays similar in energy terms but the cells are smaller.
2. I = 120,000 ÷ 384 = 312.5 A; loss = $312.5^2 \times 0.040$ = **3.91 kW**.
3. Usable energy = 58.4 × 0.8 = 46.7 kWh → 46,694 ÷ 145 = **~322 km**. Cell resistance 30 × 1.5 = 45 mΩ + 10 mΩ = 55 mΩ → loss at 390.6 A = $390.6^2 \times 0.055$ = **8.39 kW**, and sag 21.5 V.

Application

1. Ask for a breakdown: cell $/kWh against market benchmarks (BNEF 2025 LFP pack average $81/kWh); cell share of pack; enclosure material and process; thermal plate design; BMS and HV component costs; assembly cycle time and labour; test time; scrap; logistics and duties; margin.
2. 10–80% is 70% of 61.44 = 43.0 kWh into the cells. At an average of about 85 kW after taper and losses: 43.0 ÷ 85 = 0.51 h ≈ **30 minutes**, borderline. It depends on taper; with preconditioning and an LFP cell rated about 2C charge, it is achievable; in winter or with a cold pack it will not be.

16. Further learning

- Davide Andrea, *Battery Management Systems for Large Lithium-Ion Battery Packs* (Artech House).
- Warner, *The Handbook of Lithium-Ion Battery Pack Design* (Elsevier, 2nd ed.).
- UNECE Regulation No. 100 (REESS safety) and AIS-038 Rev 2.

- SAE J2464 (abuse testing) and ISO 6469 (EV safety).

Next: Module 7, the battery management system.